METRIC

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MILITARY STANDARD

MACHINERY ALTERATION (MACHALT) INSTRUCTIONS, PREPARATION OF (METRIC)

AMSC N3221

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DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND

WASHINGTON, DC 20362

Machinery Alteration (MACHALT)
Instruction, Preparation of

DOD-STD-2140(SH)

1. This Military Standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. A MACHALT instruction prepared in accordance with this standard is intended to provide personnel with all of the information necessary to effect the alteration of Hull, Mechanical and Electrical (HM&E) shipboard equipment or systems. As used in this standard, the preparer of the MACHALT instruction may be the Naval Sea Systems Command (NAVSEA) or the Naval Ship Systems Engineering Station (NAVSSES), or a contractor designated by the above.

2. The examples presented throughout this instruction are for illustrative purpose only and may not be the actual numbers assigned to the respective equipments.

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CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Scope	1
2.	REFERENCED DOCUMENTS	1
2.1	Issues of documents	1
2.2	Other publications	2
3.	DEFINITIONS	2
3.1	Machinery alteration	2
3.2	MACHALT instruction	3
3.2.1	Short form MACHALT instruction	3
3.2.2	Long form MACHALT instruction	3
3.3	MACHALT kit	3
4.	GENERAL REQUIREMENTS	3
4.1	General	3
4.2	Security classification	3
4.3	MACHALT number-revision	4
4.4	Revision of released MACHALT instruction	4
4.5	New (superseding MACHALT instruction)	4
4.6	MACHALT instruction text review and MACHALT proofing	5
4.6.1	Preliminary MACHALT instruction review	5
4.6.2	Proofing	5
4.7	MACHALT instruction approval	5
5.	DETAILED REQUIREMENTS	5
5.1	Front matter	5
5.1.1	Title page	5
5.1.1.1	Security classification	5
5.1.1.2	MACHALT instruction number and revision suffix	5
5.1.1.3	Code identification	5
5.1.1.4	Title	5
5.1.1.5	Command code	5
5.1.1.6	Special notes	6
5.1.1.7	Effective date	6
5.1.1.8	Other marking	6
5.1.2	Signature page	6
5.1.2.1	Identification of preparer	6
5.1.2.2	In-Service Engineering Activity (ISEA)	6
5.1.2.3	Government authentication	6
5.2	Feature headings and content	6
5.2.1	MACHALT intruction paragraph 1 - Distribution	7
5.2.2	MACHALT instruction paragraph 2 - Subject	7
5.2.3	MACHALT instruction paragraph 3 - Purpose	7
5.2.4	MACHALT instruction paragraph 4 - Authority	8
5.2.5	MACHALT instruction paragraph 5 - Application	8
5.2.5.1	Location or system	8
5.2.6	MACHALT instruction paragraph 6 - MACHALT accomplishment keypoint check	8

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Check the source to verify that this is the current version before use.

CONTENTS (Continued)

Paragraph		Page
5.2.7	MACHALT instruction paragraph 7 - Compliance	9
5.2.7.1	Compliance priority levels	9
5.2.8	MACHALT instruction paragraph 8 - Accomplishment categories	9
5.2.8.1	Category 1 (organization)	10
5.2.8.2	Category 2 (intermediate)	10
5.2.8.3	Category 3 (depot)	10
5.2.9	MACHALT instruction paragraph 9 - Man-hours required	10
5.2.10	MACHALT instruction paragraph 10 - Drawing references	10
5.2.10.1	Drawing availability	11
5.2.11	MACHALT instruction paragraph 11 - Enclosures	11
5.2.12	MACHALT instruction paragraph 12 - Supply data	11
5.2.12.1	Ordering data	11
5.2.12.2	Kit content	11
5.2.12.3	Special materials, tools, or test equipment required for installation and checkout but not supplied in kit	12
5.2.12.4	Special tools/support/test equipment required after installation	12
5.2.12.5	Disposition of removed materials, parts, and units	13
5.2.12.6	Alteration of spare assemblies/equipment	13
5.2.12.6.1	Assemblies/equipment in stock	13
5.2.12.6.2	Parts/materials required to modify spares	13
5.2.12.7	New/modified/additional spare parts	14
5.2.12.8	Allowance changes	14
5.2.12.8.1	Allowance parts list parts addition/allowance increase	14
5.2.12.8.2	Allowance equipage list parts addition/allowance increase	14
5.2.12.8.3	Allowance parts list parts deletion/allowance decrease	15
5.2.12.8.4	Allowance equipage list parts deletion/allowance decrease	15
5.2.13	MACHALT instruction paragraph 13 - Detailed instructions	15
5.2.13.1	Style	15
5.2.13.2	Illustrations	15
5.2.13.3	Nomenclature and part numbers	16
5.2.13.4	Markings and nameplates	16
5.2.13.5	Local stock and sources	16
5.2.13.6	Required tests	16
5.2.13.7	Descriptive terms	16
5.2.13.8	Grammatical person and mode	16
5.2.13.9	Warnings, cautions, and notes	16

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Check the source to verify that this is the current version before use.

CONTENTS (Continued)

Paragraph		Page
5.2.13.10	Abbreviations and spelling	17
5.2.14	MACHALT instruction paragraph 14 - Identification	17
5.2.14.1	Type of MACHALT plates	17
5.2.14.2	Selection of material of MACHALT plates	17
5.2.14.3	Size of MACHALT plates	17
5.2.14.4	Marking of information and format	17
5.2.14.4.1	Type of marking	17
5.2.14.4.2	Alternative materials or methods of marking	17
5.2.14.4.3	Marking	18
5.2.14.4.4	Abbreviations	18
5.2.14.5	Mounting and location	18
5.2.15	MACHALT instruction paragraph 15 - Shipping weight	18
5.2.16	MACHALT instruction paragraph 16 - Updating operational support documentation	18
5.2.16.1	Documentation identification	19
5.2.16.2	Procedure	19
5.2.16.3	Documentation availability	20
5.2.16.4	Drawing changes	20
5.2.17	MACHALT instruction paragraph 17 - Report of completion log entry	20
5.2.17.1	Nonexpendable engineering equipment	20
5.2.17.2	Expendable engineering equipment	20
5.3	Copy preparation requirements	20
5.3.1	Manner of preparation	20
5.3.2	Camera-ready copy	20
5.3.3	Illustrations	21
5.3.3.1	Location	21
5.3.3.2	Figure numbers and titles	21
5.3.3.3	Nomenclature in illustrations	21
5.3.3.4	Diagrams	21
5.3.3.5	Size consideration	22
5.3.4	Page numbering	22
5.3.5	Paragraph numbering	22
5.3.6	Figure references	22
5.3.7	Item references	22
5.3.8	Text references	22
5.3.9	Number references	22
5.3.10	Specification references	23
5.3.11	Material references	23
5.3.12	Negatives	23
5.3.13	Printing	23
6.	PACKAGING	23
6.1	Packaging	23
7.	NOTES	23
7.1	Intended use	23

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:57Z
Check the source to verify that this is the current version before use.

CONTENTS (Continued)

Paragraph		Page
7.2	Ordering data	23
7.3	Negatives and printing	24

FIGURES

Figure		
1	Example of classified MACHALT title page . . .	25
2	Example of unclassified MACHALT title page . .	26
3	Example of MACHALT revision title page	27
4	Example of MACHALT signature page	28
5	Example of MACHALT revision record sheet . . .	29
6	Example of MACHALT page (Revision authority) .	30
7	Example of MACHALT plate	31/32
8	Cutaway view of valve assembly	45

APPENDICES

Appendix		
A	Sample MACHALT instruction	33
B	Data requirements	49

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1. SCOPE

1.1 Scope. This standard covers the development, content, style, format, and general production requirements applicable to machinery alteration (MACHALT) instructions. MACHALT instructions are prepared in either short form or long form (see 3.2.1 and 3.2.2).

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting.

MILITARY

MIL-M-9868 - Microfilming of Engineering Documents, 35mm, Requirements for.
MIL-P-15024 - Plates, Tags and Band for Identification of Equipment.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements for.
MIL-P-24534 - Planned Maintenance Subsystem; Development of Maintenance Requirement Cards, Maintenance Index Pages and Associated Documentation.
MIL-M-38748 - Microfiche for Engineering/Technical Data Reports, Studies, and Related Data, Requirements for.
MIL-M-38784 - Manuals, Technical: General Style and Format Requirements.
MIL-P-38790 - Printing Production of Technical Manuals: General Requirements for.

STANDARDS

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings, Specifications, Standards and Technical Documents.
MIL-STD-130 - Identification and Marking of U.S. Military Property.
MIL-STD-143 - Standards and Specifications, Order of Precedence for the Selection of.
DOD-STD-480 - Configuration Control - Engineering Changes, Deviations, and Waivers.
MIL-STD-481 - Configuration Control - Engineering Changes, Deviations, and Waivers (Short Form).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y14.15 - Electrical and Electronic Diagrams.
Y32.2 - Graphic Symbols for Electrical and Electronics Diagrams (Including Reference Designation Class Designation Letters).
Y32.14 - Graphic Symbols for Logic Diagrams.
Y32.16 - Reference Designations for Electrical and Electronics Parts and Equipments.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

THE INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS (IPC)
IPC-T-50 - Terms and Definitions for Interconnecting and Packaging Electronic Circuits.

(Application for copies should be addressed to the Institute for Interconnecting and Packaging Electronic Circuits, 3451 Church Street, Evanston, IL 60203.)

UNITED STATES GOVERNMENT PRINTING OFFICE
Government Printing Office Style Manual.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. DEFINITIONS

3.1 Machinery Alteration (MACHALT). A Machinery Alteration (MACHALT) is any change effected on Hull, Mechanical, and Electrical (HM&E) equipments or systems that meets all of the following conditions:

(a) Can be accomplished without changing an interface external to the equipment or system.
(b) Is a modification made within the equipment boundary or is a direct replacement of the original equipment design.
(c) Can be accomplished without the ship being in an industrial activity.
(d) Will be accomplished individually and not conjunctively with a Ship Alteration (SHIPALT) or other MACHALT.

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3.2 MACHALT instruction. A MACHALT instruction is a document which contains the detailed installation and checkout instructions and other related information required to perform an alteration to an HM&E system/equipment. A MACHALT instruction will be one element of a MACHALT kit. The MACHALT instruction shall reference the other elements (provisioning data, drawings, updated support documentation, parts lists and other related information) which, are contained within a MACHALT kit.

3.2.1 Short form MACHALT instruction. A "short form" MACHALT instruction is an instruction which covers an alteration that satisfies th following criteria:

(a) The alteration does not affect the spare parts allowance of the equipment involved.
(b) The alteration does not require the furnishing of a MACHALT kit of material for accomplishment.

3.2.2 Long form MACHALT instruction. A "long form" MACHALT instruction is an instruction which does not fall within the definition in 3.2.1 (i.e., the alteration will require new or modified logistic support products or documentation and may require the furnishing of a MACHALT kit, or relate to any combination thereof).

3.3 MACHALT kit. The MACHALT kit is an assemblage of installation and checkout materials and logistic/document support products required to alter a HM&E system/equipment by accomplishment of a MACHALT. A MACHALT kit is made up of a MACHALT instruction joined with materials, drawings, updated support documentation, provisioning data and other related information required to accomplish and support a Machinery Alteration.

4. GENERAL REQUIREMENTS

4.1 General. The MACHALT instruction(s) shall be prepared in accordance with the requirements prescribed herein. The MACHALT instruction(s) shall be developed from the directive approved by the cognizant Configuration Control Board (CCB) and the requirements of Engineering Change Proposal (ECP) developed in accordance with DOD-STD-480 or MIL-STD-481. The elements of style, structure, format and content shall be as shown in figures 1 through 4, and in the sample MACHALT instruction contained in appendix A. Allowable variation of or constraints on presentation, as shown in figures 1 through 4 and the sample MACHALT instruction shall be as set forth in this standard. MACHALT instructions shall list test equipment, materials, test procedures, and documentation (including technical manuals, engineering drawings/sketches, maintenance index pages (MIPs), and maintenance requirement cards (MRCs) prepared in accordance with MIL-P-24534). Qualified personnel needed to accomplish the alteration are also cited in MACHALT instructions. For data requirements, see appendix B.

4.2 Security classification. MACHALT instructions shall be issued unclassified, whenever possible. If classified, the security classification compatible with contents shall be assigned. The security classification of the MACHALT instruction shall be assigned by authority of the

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appropriate reference security instruction cited by the acquisition authority. The security classification, if other than unclassified, shall be shown on each page (see figure 1). If a page of a classified MACHALT instruction is unclassified, the page shall so state.

4.3 MACHALT number-revision. Every page of a MACHALT instruction, original or revised, shall have the complete MACHALT identification number and the revision suffix (if any) (see figures 2 and 3). The MACHALT instruction number shall be the same as the MACHALT number assigned by NAVSEA.

4.4 Revision of released MACHALT instruction. A revision shall be a complete, new edition of an existing MACHALT instruction. Revisions shall be indicated by a capital Gothic letter immediately following the same MACHALT number as the basic MACHALT. The first revision shall be marked with the letter "A", and succeeding revisions shall be indicated by other letters in alphabetical sequence, except that the letters "I", "O", "Q", and "S", shall not be used (see figure 3). A revision shall be issued when there are changes to the technical portion of the text, with no changes to the kit content, and when no separate report of completion is required. A revision shall also be issued to correct latent defects found under contract warranty, should additions or deletions to the text or applicability be required, or to correct typographical errors. Pen and ink changes to a MACHALT instruction shall not be used. A revision shall include a revision record sheet listing all pages revised (see figure 5). Vertical black bars in the outer margins shall be used to indicate revised portions to a given page. A revision to a MACHALT instruction shall clearly bear beneath the title the legend: "Not to be accomplished on material on which MACHALT _____ revision(s) A, B, C, has (have) been completed". No revision shall be issued until after approval of the revision and identification of the appropriate revision letter by NAVSEA.

4.5 New (superseding MACHALT instruction). A new (superseding) MACHALT instruction shall be issued in lieu of a revision whenever the physical material to be supplied in the new MACHALT kit is not completely interchangeable with the material supplied in the original kit, and whenever the alteration to the equipment requires the submission of new completion reports, even though the original MACHALT installation has been performed. The new MACHALT instruction shall clearly bear beneath the title the legend: "Supersedes MACHALT _____ . To be accomplished whether or not MACHALT _____ has been performed". If the original superseded MACHALT has been performed and the accomplishment of the new MACHALT is not required, the new MACHALT shall clearly bear beneath the title the legend: "Supersedes MACHALT _____ . Not to be accomplished on material on which MACHALT _____ has been accomplished". New (superseding) MACHALT instructions shall not be prepared until after approval of a new MACHALT and identification of the new MACHALT instruction number by NAVSEA.

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4.6 MACHALT instruction text review and MACHALT proofing.

4.6.1 Preliminary MACHALT instruction review. When acquired by contract or order, MACHALT instruction shall be prepared and furnished in preliminary draft form for review by the In-Service Engineering Activity for HM&E systems and equipment (NAVSSES) (see 7.2).

4.6.2 Proofing. When acquired by contract or order, the MACHALT kit and all its contents, and related documentation, shall be proofed as being suitable for all its intended uses by the In-Service Engineering Activity for HM&E systems (NAVSSES) and equipment in accordance with the requirements of the contract or order (see 7.2).

4.7 MACHALT instruction approval. When acquired by contract or order, the material furnished in conformance with this standard shall be subject to approval by the In-Service Engineering Activity for HM&E systems and equipment (NAVSSES) (see 7.2).

5. DETAILED REQUIREMENTS

5.1 Front matter.

5.1.1 Title page. The MACHALT instruction title page shall be prepared in accordance with 5.1.1.1 through 5.1.1.8 and the format of figures 1, 2, and 3. (Type size and spacing shown in the figures are for relative size comparison and positioning only. The specific type faces indicated need not be strictly adhered to.)

5.1.1.1 Security classification. The security classification, as determined in 4.2, shall be shown in capital letters centered at the top and bottom of the page (see figure 1).

5.1.1.2 MACHALT instruction number and revision suffix. The MACHALT instruction number and revision suffix (if any) as determined in 4.3, shall be shown in capital letters at the top right-hand corner of the page (see figure 3).

5.1.1.3 Code identification. The NAVSEA design activity code identification "CODE IDENT 53711" shall be shown at the top left-hand corner of the page.

5.1.1.4 Title. The title shall completely identify the unit, equipment, subsystem and system in accordance with the official nomenclature assigned to such items by NAVSEA.

5.1.1.5 Command code. The command code shall be that of the NAVSEA life cycle manager responsible for the MACHALT (see figure 1).

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5.1.1.6 Special notes. When the detailed instruction (see 5.2.13) contains equipment or system operational procedures that require the attention of personnel who normally would not have an interest in the particular MACHALT instruction, an appropriate statement shall be placed on the title page below the command code; for example: "Bring this MACHALT instruction to the attention of all Navy personnel cleared for operation and maintenance of the (equipment)".

5.1.1.7 Effective date. The date on the title page shall be the date the MACHALT instruction is signed (see 5.1.2.3).

5.1.1.8 Other marking. The term "FOR OFFICIAL USE ONLY" shall be shown at the center of the bottom of the page.

5.1.2 Signature page. The MACHALT instruction signature page shall be prepared in accordance with 5.1.2.1 through 5.1.2.3 and as illustrated in the format of figure 4.

5.1.2.1 Identification of preparer. The name and address of the Government activity who prepared the MACHALT instruction shall be entered under the heading "PREPARED BY". MACHALT instructions delivered under contract shall also include, under this heading, the contract number and the name of any subcontractor who generated any part of such data.

5.1.2.2 In-Service Engineering Activity (ISEA). The name and address of NAVSSES shall be entered under the heading "IN-SERVICE ENGINEERING ACTIVITY FOR THE HULL, MECHANICAL, AND ELECTRICAL SHIPBOARD SYSTEM".

5.1.2.3 Government authentication. The approved MACHALT instruction will be authenticated by entering under the heading "APPROVED FOR NAVSEA", the signature, position, code and effective date of signature of the NAVSEA official to whom responsibility for such authentication has been delegated. Authentication stamps or other substitutes for a signature shall not be used.

5.2 Feature headings and content. The following paragraph headings, 1 through 17, in the order listed, shall be addressed, but may be marked "NA" or "NONE", as appropriate, when no data is required. For short form MACHALT instructions, paragraph headings 12 and 15 shall always be marked "NONE" or "NA" respectively. (Numbers in parentheses refer to relevant paragraphs in this standard.) (See sample MACHALT instruction in appendix A.)

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cal Summary Report to the reporting activity for use in reported per paragraph 4-140.

MACHALT Instruction paragraph	Title
1.	Distribution (see 5.2.1)
2.	Subject (see 5.2.2)
3.	Purpose (see 5.2.3)
4.	Authority (see 5.2.4)
5.	Application (see 5.2.5)
6.	MACHALT accomplishment keypoint check (see 5.2.6)
7.	Compliance (see 5.2.7)
8.	Accomplishment categories (see 5.2.8)
9.	Man-hours required (see 5.2.9)
10.	Drawing references (see 5.2.10)
11.	Enclosures (see 5.2.11)
12.	Supply data (see 5.2.12)
13.	Detailed instructions (see 5.2.13)
14.	Identification (see 5.2.14)
15.	Shipping weight (see 5.2.15)
16.	Updating operational support documentation (see 5.2.16)
17.	Report of completion log entry (see 5.2.17)

5.2.1 MACHALT instruction paragraph 1 - Distribution. A distribution list shall be included as part of the instruction. The distribution list will be provided by the activity having technical authority at the time of approval of the draft MACHALT instruction submitted for review. The list shall include the activities to whom copies of the MACHALT instructions are to be issued. These activities shall be designated by symbols specified in the most current edition of the Standard Navy Distribution List. The distribution list shall also contain the symbols for all applicable ships and stations and such fleet, force, type, and other commands, repair ships and facilities, etc., as may be appropriate (depending on the nature of the MACHALT) or as may be directed by NAVSEA. A footnote to the distribution list shall be included stating the repository where additional copies of the MACHALT instruction may be obtained.

5.2.2 MACHALT instruction paragraph 2 - Subject. This paragraph shall identify the component, etc., and as applicable, the next higher assembly and the system. A summary of the action required shall follow. Serial numbers shall be included for complete identification whenever a limited series of a particular equipment is to be altered by the specific MACHALT kit. Approved nomenclature and system application shall be used. The phrase "All Mods" shall not be used as part of the identification or description of the equipment.

5.2.3 MACHALT instruction paragraph 3 - Purpose. The text shall provide a clear and concise functional description of the alteration, with explanations of the reasons and advantages. When the alteration is required to meet a specific HM&E system or equipment capability, the MACHALT text shall so state. When the MACHALT instruction being prepared is a revision to a existing MACHALT instruction, a subheading shall be inserted and titled "Purpose of Revision". The subparagraph shall include the reason for and the scope of the revision.

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5.2.4 MACHALT instruction paragraph 4 - Authority. This paragraph shall list the NAVSEA Configuration Control Board (CCB) Directive number and date and other documents authorizing preparation of the MACHALT instruction. This paragraph shall also include authorization for any revised MACHALT instruction (see figure 6). For each Engineering Change Proposal (ECP) involved in the MACHALT instruction, the ECP number, the NAVSEA Navy Control Number (NCN), and the originator's Unit Identification Code (UIC) shall be included.

5.2.5 MACHALT instruction paragraph 5 - Application. The specific HM&E system or equipment to which the MACHALT instruction applies shall be identified by the most complete nomenclature (e.g., system, subsystem, equipment, model number, serial number). The phrase "All Mods" shall not be used. Serial numbers or ship and hull numbers shall be included; for example:

"This MACHALT instruction is applicable to (equipments), model (number), serial (number), installed or in store. The equivalent of this MACHALT was accomplished during production on equipments having serial numbers higher than (number)."

or

"This MACHALT instruction is applicable to (equipment), model (number), serial number (list) to (list) inclusive."

5.2.5.1 Location or system. When the MACHALT instruction is applicable only to equipment installed in a specific location, ship, ship class, or system configuration, this applicability should be so stated; for example:

"This MACHALT instruction is applicable to the starboard forward-most (equipment or system) on destroyers with hull numbers (list)."

or

"This MACHALT instruction is applicable to (equipment or system) on the following ships only:

NAME/HULL NUMBER NAME/HULL NUMBER NAME/HULL NUMBER"

5.2.6 MACHALT instruction paragraph 6 - MACHALT accomplishment keypoint check. This paragraph shall include a complete MACHALT verification list that identifies physical and functional features of the equipment in sufficient detail to enable operating personnel to determine with certainty the accomplishment status of this MACHALT.

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5.2.7 MACHALT instruction paragraph 7 - Compliance. This paragraph shall describe how the MACHALT shall be accomplished on all applicable equipments and the priorities based on this "how"; for example:

"This MACHALT shall be incorporated by intermediate level maintenance activity personnel on those (equipments or systems) identified in paragraph 5 that are installed on operating ships that will not be under SCN funding at the time of MACHALT installation. Installations shall be in accordance with the installation priorities established by the CNO Amalgamated Military/Technical Improvement Plan (AMT) based on an assigned OPNAV MACHALT priority Level #4, Routine Combat/Primary System Modernization. For those (equipments or systems) identified in paragraph 5 that are installed or to be installed on ships under SCN funding: this MACHALT will be incorporated during the production of each (equipment or system) on a schedule in concert with its manufacturing process or by intermediate level maintenance activity personnel in accordance with an installation schedule established by the shipbuilders."

or

"The intent of this MACHALT shall be incorporated during the production on each (equipment or system) Serial Nos. (list) and higher on a schedule in concert with its manufacturing process and by depot level maintenance personnel on (equipment or system) Serial Nos. (list) and lower on a schedule in concert with each (equipment or system) scheduled location at the depot level maintenance activity for repair or overhaul."

5.2.7.1 Compliance priority levels. The MACHALT priority code to be assigned to MACHALT applicable to installed equipments on operating ships not under SCN funding, shall be as specified by the directive approved by the cognizant Configuration Control Board (CCB). Alteration priority levels are identified as follows:

(a) Mandatory and safety.
(b) Reliability and maintainability (primary).
(c) Urgent combat/primary system modernization.
(d) Routine combat/primary system modernization.
(e) Reliability and maintainability (secondary).
(f) Priority secondary mission area modernization.
(g) Routine secondary mission and modernization.
(h) Mission/reliability support.

5.2.8 MACHALT instruction paragraph 8 - Accomplishment categories. Accomplishment categories shall be based on the maintenance level and skills required to accomplish the MACHALT.

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5.2.8.1 Category 1 (organization). Category 1 shall include those MACHALT approved for accomplishment by custodial (ship or activity) or technical-installation-team personnel qualified to perform the installation.

5.2.8.2 Category 2 (intermediate). Category 2 shall include those MACHALT approved for accomplishment by intermediate maintenance activities (e.g., tenders, NAVSEACENs, MOTUs, SIMAs).

5.2.8.3 Category 3 (depot). Category 3 shall include those MACHALT approved for accomplishment by an assigned depot level maintenance activity. The installed equipment is removed from the ship and transferred to the depot level maintenance activity for equipment refurbishment and MACHALT installation. These MACHALT require equipment and skills corresponding to those required for depot level maintenance. Category 3 also shall include those MACHALT approved for accomplishment by production contractor or by a team of qualified personnel.

5.2.9 MACHALT instruction paragraph 9 - Man-hours required. This paragraph shall provide total man-hours required for removal, alteration, reinstallation, and testing of the equipment affected the the alteration. Actual man-hours determined from experience or from trial alterations, shall be furnished whenever possible. Final texts shall reflect man-hour adjustments based on accomplishment of shipboard prove-in. The most accurate information shall be made available to support planning, budgeting, and programming of MACHALTs.

5.2.10 MACHALT instruction paragraph 10 - Drawing references. These documents shall be listed by NAVSEA or DOD identification numbers, the latest revision letters and titles. Contractor's numbers may be used when no NAVSEA or DOD numbers are available. Drawings shall be identified by Federal Supply Code for Manufacturers (FSCM) number in parentheses. NAVSEA drawings shall have the applicable code identification followed by the drawing number. Drawings and associated lists, or sheets thereof, that are deleted by the MACHALT from the set of drawings applicable to the specific equipment to which the MACHALT applies shall be listed. Instructions for disposal, if necessary, shall be included. Information shall note whether the drawings are still applicable to other equipment in the system or in related shipboard systems. In addition, drawings that were revised in the set applicable to the specific equipment and any new drawings and lists that were generated to support the MACHALT kit shall be listed. Drawings shall be listed in the following order:

(a) New drawings generated and required for the accomplishment of the MACHALT (supplied as part of the MACHALT kit).
(b) Drawings which need to be revised due to the accomplishment of the MACHALT. This list shall state source for obtaining.
(c) Obsolete drawings deleted by accomplishment of the MACHALT. This list shall state disposition for deleted drawings.
(d) Drawings and references for information.

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5.2.10.1 Drawing availability. Aperature cards (or other microform) for new drawings to support the altered equipment shall be made available by the acquisition manager as part of the MACHALT kit. Aperature cards for new drawings shall also be forwarded to the planning yard and appropriate PERA for the installation ship for updating the appropriate plan files and ships information book after installation is complete. Aperature cards (or other microform) for revised drawings to support altered equipment will be separately supplied by delegated agents. The source of such copies shall be provided so they may be requisitioned should the normal distribution system fail to supply them.

5.2.11 MACHALT instruction paragraph 11 - Enclosures. This paragraph shall list all required data supplied as enclosures to the MACHALT instruction (e.g., test specifications, service bulletins, etc.). Tear-out sheet(s) shall be provided for interim Allowance Parts List (APL) and Allowance Equipage List (AEL) support. The information required in 5.2.12.8 shall be duplicated for that purpose.

5.2.12 MACHALT instruction paragraph 12 - Supply data. This paragraph shall list the following: parts and materials supplied in the MACHALT kit; parts, materials, test equipment, and special tools required for the alteration but not supplied in the kit; spare assemblies/equipment requiring modification and the parts, materials, and special tools required to modify those spares; parts and material removed from expendable, installed, or spare equipment and their disposition; and the source of kits, parts, materials, test equipment, and special tools. If supply items are not required, the word "None" shall be entered after the feature heading and subheadings omitted. If supply items are required, compliance with 5.2.12.1 through 5.2.12.8 shall be required. The column headings "National Stock Number" and "Source, Maintenance Recoverability (SM&R) Code" shall be shown even though no data are entered in these columns. When a part that is listed does not have a Government assigned/controlled number, the manufacturer's part number shall be listed. Part numbers shall be preceded with a five-digit FSCM code number; for example: (06848) 13026-3.

5.2.12.1 Ordering data. Ordering data shall include the issuing authority (i.e., ISEA) from whom the MACHALT kit must be ordered for items in the kit and the kit number assigned.

5.2.12.2 Kit content. All elements contained within a MACHALT kit shall be listed in the MACHALT instruction. Materials, special tools, special test equipment, Coordinated Shipboard Allowance List (COSAL) repair parts (if appropriate), updated support documentation, and new drawings required for MACHALT accomplishment and supplied in the MACHALT kit shall be listed. Each item in the kit shall be labeled to identify its item number, nomenclature, and part number. Special tools supplied in the kit shall include those newly acquired to repair and maintain the equipment after completion of the MACHALT, plus special tools required to complete MACHALT installation. It shall be mandatory that an initial quantity of unique, modified, or substituted repair parts that will become organizational-level-of-repair spare items, and that are not currently included in the organizational-maintenance-level allowances, be included in

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the MACHALT kit. Such items shall be limited to part-numbered items and items approved by the Inventory Control Point (ICP). The Military Essentiality Codes (MECs) and SM&R codes shall be supplied for all such items included in the kit. Complete supply nomenclature and full instructions for establishing permanent COSAL allowance quantities shall also be included. Information shall be provided in tabular form using the following column headings:

Item No.	Nomenclature/ Description	National Stock No.	FSCM	Part No.	MEC Code	SM&R Code	MACHALT Kit Quantity

5.2.12.3 Special materials, tools, or test equipment required for installation and checkout but not supplied in kit. A tabular listing of all special materials, tools, or test equipment required for accomplishment, but which are not supplied in the kit, shall be provided using the headings shown below. Common tools carried in the tool box need not be mentioned. Tabular listing shall be as follows:

Nomenclature/ Description	National Stock No.	FSCM	Part No.	Quantity Required	Stocking Activity/ Source	Remarks

The "Stocking Activity/Source" column shall indicate where nonsupplied items can be obtained. If parts from ship's stock are to be used, the "Remarks" column shall indicate the number of those parts that should be ordered for replenishment (including backup spares). Parts or materials that are required to comply with the instructions but are not practical to include in the kit due to packaging, storing, shipping, or processing difficulties (e.g., paint, abrasive paper, cement, oil, explosive rivets, soldering or welding material) shall be listed. When listing parts or materials such as sheet stock, overall dimensions (not the area) shall be indicated in the "Quantity" column. If the material is to be used to alter more than one item, it shall be so stated. When an exact amount cannot be given, an estimate of volume per equipment shall be provided; for example: "approximately 1 pint". The term "as required" shall be avoided.

5.2.12.4 Special tools/support/test equipment required after installation. The special tools, support equipment, and test equipment (with required calibration requirements) required for equipment operation or maintenance after MACHALT installation shall be listed in tabular form using the headings shown below. The test, measurements, and diagnostic equipment, including Special Purpose Test Equipment (SPTE) and General Purpose Test Equipment (GPTE), applicable subcategory (SCAT) code and related ancillary equipment and accessories shall be included. Any advice or instructions regarding changes in the Ship's Portable Electrical/ Electronic Test Equipment Requirement List (SPETERL) shall be entered under the "Remarks" column. Specific instructions for adjustment of ship's COSAL shall also be included under "Remarks" to reflect such additional special tools, test equipment, or support equipment. Tabular headings shall be as follows:

Nomenclature/ Description	National Stock No.	FSCM	Part No.	Quantity	Stocking Activity	SCAT Code	Remarks

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5.2.12.5 Disposition of removed materials, parts, and units. All parts or materials that are removed from installed or spare equipment shall be listed. Quantity, part number, nomenclature, NSN, and disposition code shall be indicated. Parts removed for rework and subsequent reinstallation or for use as spares, regardless of where the rework is to be performed, shall also be included. Any unusual conditions associated with the removal of parts or materials shall be explained. Removed parts and materials shall be listed under column headings as follows:

Quantity	Part No.	Nomenclature	National Stock No.[1/]	Disposition Code

If no parts are removed by the alteration, the word "None" shall be entered after the subhead title and the column headings omitted. The following standard disposition code numbers and their explanations shall be used (include only those applicable in numerical sequence):

Code	Explanation
1	Disposition of replaced material is at discretion of Commanding Officer.
2	Scrap.
3	Rework in accordance with detailed instructions.
4	Return to stock if repairable or serviceable.
5	Use until stock is exhausted.
6	Return item to contractor.
7	Use additional as necessary.

5.2.12.6 Alteration of spare assemblies/equipment.

5.2.12.6.1 Assemblies/equipment in stock. The spare assemblies/equipment (ship, tender, or Navy supply system) to which the MACHALT will be applicable shall be furnished under column headings as follows:

Model No.[2/]	Nomenclature	National Stock No.	Control Number[2/]	Serial Number[2/]	Location of Spares

If no modification or spares is required, the word "None" shall be entered after the subheading title and the column headings omitted.

5.2.12.6.2 Parts/materials required to modify spares. The parts or materials required to accomplish the MACHALT in spare assemblies/equipment shall be specified using the headings and arrangements in 5.2.12.2. Introductory statements preceding these listings may be modified as necessary. When the parts or material required to modify spares are the same as those listed in 5.2.12.2 or 5.2.12.3, a statement so indicating may be used instead of repeating the listing. If neither parts nor materials are required, the word "None" shall be entered after the subhead title.

[1/] Nonstock items removed and designated for "scrap" shall not be stock numbered.
[2/] Omit these columns if not applicable.

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5.2.12.7 New/modified/additional spare parts. If on board spare assemblies are not to be altered concurrent with the system/equipment, new or modified spare assemblies shall be supplied with the MACHALT kit. The spare assemblies that are to be supplied shall be identified in tabular form as follows:

Item No.	Nomenclature/ Description	National Stock No.	Part No.	Quantity

If no new or modified spare assemblies are required in support of the altered equipment, the word "None" shall be entered after the subheading and the column headings omitted.

5.2.12.8 Allowance changes. Items that will be added to or deleted from the allowance lists as a result of the equipment alteration shall be listed. Note that parts population can change without affecting the allowance quantity. These lists shall be included in an enclosure to the MACHALT instruction so that they can be removed separately and attached to the affected APL/AEL. Repair parts or spares furnished with the MACHALT kit as allowance items shall be designated in 5.2.12.8.1 and 5.2.12.8.2 by use of a double asterisk (**) and so explained in a footnote. Entries in 5.2.12.8.1 and 5.2.12.8.3 which indicate a parts population change only and which do not affect allowance quantities shall be annotated by an asterisk (*) and so explained in a footnote.

5.2.12.8.1 Allowance parts list parts addition/allowance increase. The following items/quantities are to be added to the present APL after equipment is altered:

ALLOWANCE PARTS LIST NO.

Reference Designator	Part No.	Nomenclature	National Stock No.	SM&R Code	Parts Population Added	Allowance Quantity Increase

5.2.12.8.2 Allowance equipage list parts addition/allowance increase. The following items/quantities are to be added to the present AEL after equipment is altered:

ALLOWANCE EQUIPAGE LIST NO.

Part No.	Nomenclature	National Stock No.	SM&R Code	Allowance Quantity Increase

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5.2.12.8.3 Allowance parts list parts deletion/allowance decrease. The following items/quantities are to be deleted from the present APL after equipment is altered:

ALLOWANCE
PARTS
LIST NO.

Reference Designator	Part No.	Nomenclature	National Stock No.	Parts Population Decrease	Allowance Quantity Decrease

5.2.12.8.4 Allowance equipage list parts deletion/allowance decrease. The following items/quantities are to be deleted from the present AEL after the equipment is altered:

ALLOWANCE
EQUIPAGE
LIST NO.

Part No.	Nomenclature	National Stock No.	Allowance Quantity Decrease

If no item/quantity change is required, the word "None" shall be entered after the subhead title and the column headings omitted.

5.2.13 MACHALT instruction paragraph 13 - Detailed instructions. The material furnished with the kit shall be physically and functionally compatible with the latest configuration of the shipboard system (basic or as modified by other MACHALT instructions) and with MACHALTs affecting related equipment components. The instruction shall be detailed and shall cover, as necessary, disassembly, alterations, reassembly, tests, precautions, and markings.

5.2.13.1 Style. Clarity and completeness are essential in the writing of MACHALT instructions. Extraneous information shall not be included. As many subparagraphs as necessary shall be used for step-by-step procedures, in chronological sequence, to facilitate accomplishment of the MACHALT in the most direct and simple manner consistent with personnel and equipment safety. The chronological, step-by-step procedures shall describe the equipment or system alteration and the necessary alterations to spares and replacement units.

5.2.13.2 Illustrations. Detailed instructions shall be supported by illustrations, as necessary, to supplement engineering drawings. Alterations, accomplishment of major operations in proper sequence, and installation tools in use shall be shown. The illustrations may be perspective drawings, orthographic drawings, photographs, etc. Illustrations and diagrams shall be in accordance with 5.3.3.

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5.2.13.3 Nomenclature and part numbers. Reference to items (at all levels of breakdown) shall be by both nomenclature and part number. Dash numbers from part drawing shall be used for specificity when referring to parts in text. Complete nomenclature of equipment or material shall be used for identification.

5.2.13.4 Markings and nameplates. Detailed instructions for the placing of identification markings, nameplates, shall be in accordance with MIL-STD-130. This shall be in addition to the identification of MACHALT accomplishment (i.e., stamping of MACHALT plate) required by 5.2.14. This shall be done by reidentifying the modified subassembly or component with the proper revision letter, or new assembly part number.

5.2.13.5 Local stock and sources. The use of local stock or the requirement for local manufacture of parts shall be specified only when such procedure is considered practicable to expedite incorporation of the MACHALT, and only when prior approval has been obtained from the ISEA for the MACHALT kit.

5.2.13.6 Required tests. As necessary, tests shall be specified for proper operation of the affected parts and systems. When equipments modified by MACHALT have multisystem application, and different test procedures apply to each system, the test requirements shall be grouped and properly categorized by system application.

5.2.13.7 Descriptive terms. Repetition of words, phrases, numbers, etc., shall be avoided except when required for clarity. When repetitions are necessary they shall be consistent: for example, part nomenclature from part drawings shall be used.

5.2.13.8 Grammatical person and mode. The second person imperative shall be used for operational procedures; for example: "Break casing bead loose from wheel flange." The third person shall be used for description and discussion; for example: "The torsion link assembly transmits torsional loads from the axle to the shock strut."

5.2.13.9 Warnings, cautions, and notes. Warnings, cautions, and notes shall be used as defined herein. The individual words "Warning", "Caution", and "Notes" shall appear in bold capital letters above or preceding the descriptive precautionary condition. Warnings and cautions shall precede the instructions for the procedure or practice involved. Notes may precede or follow the applicable text. Warnings, cautions, and notes shall not receive a paragraph number. Warnings, cautions, and notes which are a part of the text shall be defined as follows:

(a) WARNING: An operating procedure or practice that will result in personal injury, loss of life, or destruction of equipment if not carefully followed.
(b) CAUTION: An operating procedure or practice that will result in damage to equipment if not strictly observed.
(c) NOTE: An operating procedure or condition that must be emphasized. (This provision should be exercised judiciously).

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5.2.13.10 Abbreviations and spelling. Abbreviations shall be in accordance with MIL-STD-12 and the Government Printing Office (GPO) Style Manual. Spelling shall be in accordance with the GPO Style Manual.

5.2.14 MACHALT instruction paragraph 14 - Identification. When reidentification of an equipment or component parts is accomplished as a result of the incorporation or compliance with a MACHALT instruction, the MACHALT number and the original basic serial number shall be stamped on the MACHALT plate to be affixed to each equipment having a name plate and being modified by the MACHALT instruction. The method and format of stamping the MACHALT number on the identification plate shall be identified in the MACHALT instruction in accordance with figure 7 and 5.2.14.1 through 5.2.14.5 herein.

5.2.14.1 Type of MACHALT plates. MACHALT plates shall be furnished in accordance with MIL-P-15024, types A, B, or C.

5.2.14.2 Selection of material of MACHALT plates. Material for MACHALT plates shall be in accordance with MIL-P-15024. MACHALT plates shall be made of material that shall withstand the same environmental conditions as the item to which the plates will be attached. MACHALT plates shall be of the same material as that of the item to which the plate is installed. When MACHALT plates are to be attached to materials other than steel, brass, and aluminum, plates and screws shall be corrosion-resisting steel in accordance with QQ-S-766.

5.2.14.3 Size of MACHALT plates. Size of MACHALT plates shall be in accordance with figure 7 or a reasonable facsimile sized in accordance with MIL-P-15024.

5.2.14.4 Marking information and format. Marking information to appear on MACHALT plates and the format together with the desired special characteristics shall be in accordance with the applicable figure included herein. The marking information for a particular item shall be in accordance with the applicable MACHALT instruction.

5.2.14.4.1 Type of marking. Letters shall be of Gothic capitals and the numerals shall be Arabic. Other characters shall be of similar appearance. The Government nomenclature (name and Government designation of the item to which the nameplate will be attached) shall be emphasized by the use of larger letters.

5.2.14.4.2 Alternative materials or methods of marking. The use of alternative materials or methods of marking shall be specifically approved by the ISEA. Plastic nameplates shall not be used. If alternative marking is required, the type and material of MACHALT plate shall be in accordance with MIL-P-15024.

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5.2.14.4.3 Marking. MACHALT plates shall include the following information as specified:

(a) Name of item.
(b) Part number.
(c) Serial number.
(d) Manufacturer's name.
(e) MACHALT number.

5.2.14.4.4 Abbreviations. The format of figure 7 is considered typical. Item names and manufacturer's names vary in length causing critical limitations in available space on smaller nameplates. Abbreviations conforming to MIL-STD-12 may be used where space is limited. Manufacturer's and contractor's Federal Supply Code Number may be used in lieu of manufacturer's name. Numbers may be omitted from terms such as serial number, part number, and contract number where space is limited.

5.2.14.5 Mounting and location. The MACHALT instruction shall identify the method for mounting and location of the MACHALT plate. MACHALT plates shall be located as close as possible to the nameplate. Systems or set nameplates shall be mounted to the principal or most prominent item of the major assembly. Plates shall be located in locations easily accessible during operation. The mounting and location of the plate shall be shown on the assembly drawing of the item. Plates shall not be positioned so as to interfere with the controls or obscure other required information. The mounting and location of plates shall not have an adverse effect on the strength of the item on which the plate is mounted. In general, plates shall not be mounted by means of rivets, self-tapping screws or welding. MACHALT plates shall be attached by removable type screws (made of the same material as the nameplate) or by lockwire method.

5.2.15 MACHALT instruction paragraph 15 - Shipping weight. Shipping weight per package and the total weight and volume of the MACHALT kit shall be provided and expressed in kilograms (pounds) and cubic meters (cubic feet).

5.2.16 MACHALT instruction paragraph 16 - Updating operational support documentation. Publications and fleet-support documentation (with the exception of drawings) affected by the MACHALT, including existing technical documents, maintenance requirement cards, and allowance parts lists, etc., shall be identified and listed in the following sequence:

(a) Technical manuals.
(b) Maintenance index pages/maintenance requirement cards (MIPs/MRCs).
(c) Allowance lists (APLs, AELs).
(d) Equipment identification codes (EICs).
(e) Technical repair standards (TRSs).
(f) Total ship test procedures (TSTPs).
(g) Engineering operation sequencing system (EOSS).
(h) Other data (to include training documentation (if appropriate)).

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5.2.16.1 Documentation identification. Technical manuals that will support the equipment as altered shall be identified by publication number, volume, title, and revision/change status. In response to emergency needs, Advance Change Notices (ACNs) may be prepared and issued prior to permanent changes and revisions. MIPs and MRCs shall be identified by control number and title. APLs shall include the MACHALT APL number and all applicable parent APL numbers.

5.2.16.2 Procedure. If no documents are affected, the word "None" shall be entered after the feature heading. If any of the publication subheadings (i.e., technical manuals, MRCs, etc.) are not affected by the MACHALT, the word "None" shall be inserted after the appropriate subheading. An example of presentation is as follows:

"16. UPDATING OPERATIONAL SUPPORT DOCUMENTATION

16.1 The following publication changes are required for system/equipment/support following MACHALT accomplishment.

16.1.1 Technical manual.

16.1.1.1 NAVSEA 0951-LP-038-6020

16.1.2 Maintenance Index Page/Requirement Card (MIP/MRC).

16.1.2.1 MIP WS-61/174-C1, MRC CO-2UGU-N

16.1.3 Allowance Parts List (APL)

16.1.3.1 APL 7864034

16.1.3.2 Interim support of this MACHALT is provided by MACHALT APL 6224001. When all installations are complete, the parent APL listed in 17.1.3.1 will be updated in accordance with paragraph 12.8.

16.1.4 Equipment Identification Code (EIC).

16.1.4.1 (EIC) GB43.

16.1.5 Technical Repair Standards (TRSs).

16.1.5.1 None

16.1.6 Total Ship Test Procedures (TSTPs).

16.1.6.1 3B261C301

16.1.7 Engineering Operational Sequencing System (EOSS).

16.1.7.1 None

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16.1.8 Other data.

16.1.8.1 None."

5.2.16.3 Documentation availability. The updated support documentation specified in 5.2.16 shall be made available by the acquisition manager as part of the MACHALT kit. Automatic distribution of subsequent maintenance documentation to the Command will be accomplished by submission of form OPNAV 4790/CK, as set forth in 5.2.17.1, by MACHALT installer. The instructions shall also state that this material shall not be used to support the effected equipment/system until the MACHALT is accomplished.

5.2.16.4 Drawing changes. Drawings required to support the MACHALT kit are specified in 5.2.10.

5.2.17 MACHALT instruction paragraph 17 - Report of completion log entry. Each MACHALT instruction shall include a statement delineating the report of completion and log entry requirements.

5.2.17.1 Nonexpendable engineering equipment. Text shall direct that accomplishment of a MACHALT shall be reported to the Maintenance Data Collection System of the 3-M Program by the Maintenance Data Form OPNAV 4790/CK. EIC numbers which will be used for MACHALT accomplishment reporting on Form OPNAV 4790/CK shall be listed. The activities (including the Life Cycle Manager (LCM), the In-Service Engineering Activity (ISEA) and the Inventory Control Point (ICP)) to whom a report of accomplishment will be submitted shall be indicated.

5.2.17.2 Expendable engineering equipment. Text shall direct that accomplishment of a MACHALT shall be recorded in the prescribed log or record book. The activities (including the Life Cycle Manager (LCM), In-Service Engineering Activity (ISEA), and the Inventory Control Point (ICP)) to whom a report of accomplishment will be submitted shall be indicated. Specific details to be reported shall be included.

5.3 Copy preparation requirements.

5.3.1 Manner of preparation. When specified in the contract or order (see 7.2), the approved MACHALT instruction shall be furnished in camera-ready copy form. Except as otherwise specified herein, the camera-ready copy shall be in accordance with MIL-M-38784. In addition, the copy shall be satisfactory for use in making 35 millimeter (mm) microfilm of each page in accordance with legibility paragraph of MIL-M-9868 or microfiche in accordance with MIL-M-38748.

5.3.2 Camera-ready copy. The text shall be prepared in a single column format with left margin justification only. Except for certain presentations authorized by the contracting activity, the page size shall be approximately 21.59 cm (8.5 inches) by 27.94 cm (11 inches), with image area restricted to 17.78 cm (7 inches) by 25.4 cm (10 inches) including running head and foot. Left margin shall be 3.175 cm (1.25 inches) wide. Equipment used for preparing the instruction and marginal copy shall be

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adequate for the purpose intended and shall provide clearly legible easily usable publications at the most economical cost. Suggested type styles and sizes shall be in accordance with MIL-M-38784. Minimum acceptable material shall have the following features:

(a) Single spacing.
(b) Unjustified right margin.
(c) Headings prepared on the same composing equipment as the text.

5.3.3 Illustrations. Illustrations shall be either in the form of line drawings, continuous-tone original art, or glossy photographic prints (unscreened) intended for half-tone reproduction in accordance with MIL-M-38784. Color illustrations and cartoons shall not be used. Line drawings prepared for assembly into oversize sheets shall be reduced reproduction size without loss of detail and without closing up. Reproduction size line drawings shall be mounted in position.

5.3.3.1 Location. Illustrations and tables shall be located as close to the related text as possible or as enclosures to the MACHALT text, whichever is deemed most advantageous for clarity. Illustrations less than page size may be grouped and presented on a single page, with a figure number and title assigned to each illustration. Grouped illustrations, with one figure number and title and separate subtitles, are particularly suitable for illustrating sequential operations.

5.3.3.2 Figure numbers and titles. Illustrations shall be identified by a figure number and title. Figure numbers shall be assigned consecutive Arabic numbers. Figure numbers and titles shall be typed as part of the text, not a part of the illustration. Illustrations requiring more than one page shall be identified in the following manner:

Figure _____ . Installation Drawing (Sheet 1 of 2)
Figure _____ . Installation Drawing (Sheet 2 of 2)

5.3.3.3 Nomenclature in illustrations. Significant features or components of illustrations shall be identified by index numbers or brief nomenclature. When an exploded-view drawing involves relatively few parts, the nomenclature for each part shall be used. When many parts are involved, the parts shall be numbered, preferably clockwise, starting at the upper left, and a key list added to show the nomenclature of each numbered part. Nomenclature of parts on illustrations shall be consistent with that in the text.

5.3.3.4 Diagrams. Component parts of wiring and schematic diagrams shall be symbolized in accordance with ANSI Y32.2, ANSI Y32.14, and IPC-T-50. Reference designations shall be in accordance with ANSI Y32.16 and ANSI Y14.15. Wiring and schematic diagrams shall be planned and arranged for page-size presentation, whenever possible. Foldout pages shall be limited to two-page foldouts. In no case shall they exceed an

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overall length of 114.3 cm (45 inches) from the binding edge, with a maximum printing area of 110.49 cm (43.5 inches) long by 25.4 cm (10 inches) high. Foldout pages conforming to the requirements of this standard and the paper stock for printed manuals paragraph of MIL-M-15071 shall be used.

5.3.3.5 Size consideration. When an illustration is of such size that it cannot be legibly reproduced within the maximum single-page printing area, it may be arranged for reproduction as a foldout. Illustrations requiring more than one foldout shall be adequately identified at the breaking points to assist the reader in making the crossover.

5.3.4 Page numbering. The first or title page shall not be numbered. The signature page shall be lower case Roman number (i.e., ii). The pages following the signature page shall be numbered consecutively commencing with "1". Page numbers shall be placed at the bottom center of each page.

5.3.5 Paragraph numbering. Feature paragraph headings shall be numbered as shown in 5.2. Specification-style subparagraphing, as shown in appendix A of this standard, shall be used.

5.3.6 Figure references. References shall be to figure numbers, not page numbers. Reference to indexed items shall include both index and figure number; for example, "The 'ON-OFF' switch (34, figure 3)..." References in the same paragraph to several items in the same figure (as in instructions detailing work procedures) shall include the figure number once in that paragraph; for example:

"(a) Disassemble Air Valve (see figure 6). Unscrew safety-disc retainer (2) from valve body (1) and remove safety-disc (3) and safety-disc washer (4)."

or

"(a) Unscrew safety-disc retainer (2, figure 6) from valve body and remove safety-disc (3) and safety-disc washer (4)."

5.3.7 Item references. When referring to components of the MACHALT kit, the item number (see 5.2.12.2) shall be used. When referring to a control, the nameplate portion of the control's description shall be capitalized; for example: "POWER ON/OFF switch" or "DAMPER CONTROL".

5.3.8 Text references. When referring to text matter placed elsewhere in the instruction, reference shall be made to the paragraph and not to the page number.

5.3.9 Number references. Numbers less than 10 shall be spelled out; 10 and over shall be Arabic numerals. Double statement of numbers (a number spelled out followed by a numeral in parentheses) shall not be used. Decimal numbers less than a unit shall begin with zero before the decimal point.

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5.3.10 Specification references. When Government specification numbers are referenced, the basic number shall be listed without the revision letter suffix, unless absolutely essential to include the suffix because of major differences in the revised specification. Reference to specifications shall be made in the following manner: "Military Specification MIL-C-5020", "Federal Specification QQ-B-626", etc.

5.3.11 Material references. Materials necessary to accomplish the MACHALT (e.g., lubricants, sealing materials, abrasives, etc.) shall be identified by references to applicable specifications and standards. The applicable type, grade, class, condition, etc., shall be indicated. Where identical materials are identified by more than one document, the material shall be referenced in the order of precedence specified by MIL-STD-143. When materials are required which cannot be identified adequately by references as indicated above, additional information for complete identification shall be provided and shall include the following:

(a) Trade name(s) or commercial designations.
(b) Name and address of the producer of the materials or his FSCM number.
(c) Chemical composition (where applicable).
(d) Physical and mechanical properties in sufficient detail to disclose strength and safety characteristics when required by the design.
(e) Dielectric properties for electrical insulating materials.

5.3.12 Negatives. When specified in the contract or order (see 7.2), photolithographic negatives of the approved MACHALT instructions shall be provided in accordance with MIL-P-38790.

5.3.13 Printing. When specified in the contract or order (see 7.2), printed copies of the approved MACHALT instructions shall be provided in accordance with MIL-P-38790.

6. PACKAGING

6.1 Packaging. Packaging of camera-ready copy and related artwork for shipment shall be in accordance with MIL-M-38784. The packaging of negatives and printed copies shall be in accordance with MIL-P-38790.

7. NOTES

7.1 Intended use. This standard is intended for use in preparing MACHALT instructions issued by NAVSEA.

7.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this standard.
(b) Form(s) of MACHALT instruction(s) required (see 3.2.1, 3.2.2 and 4.1).
(c) When preliminary draft shall be provided (see 4.6.1).

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(d) Organization(s) or activity(ies) responsible for review and approval of MACHALT instruction for completeness, accuracy, and compliance with this standard (see 4.6.1, 4.6.2, and 4.7).
(e) Organization(s) and activity(ies) responsible for planning and for accomplishment of the proofing of the MACHALT kit hardware and affected documentation to test their effectiveness for intended uses (see 4.7).
(f) Compliance priority code assigned to the MACHALT (see 5.2.7.1), if applicable to nonexpendable engineering and planned for accomplishment on operating ships not under SCN funding.
(g) When a camera-ready copy is required (see 5.3.1).
(h) When photolithographic negatives are required (see 5.3.12 and 7.3).
(i) When printed copies are required (see 5.3.13 and 7.3).

7.3 Negatives and printing. Normally, the printing of approved MACHALT instructions will be accomplished by the Naval Publication Printing and Service Office. The acquisition of printed production items (i.e., photolithographic negatives, printed copies, etc.) shall be in accordance with the Department of Navy Publications and Printing Regulation (NPPR) P-35. Such acquisition shall be authorized by a Joint Committee on Printing Waiver (JCP 23383) as indicated in NPPR P-35.

Preparing activity:
Navy - SH
(Project CMAN-N010)

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CODE IDENT 53711 CONFIDENTIAL MACHALT_____

GROUP 3
DOWNGRADED AT 12-YEAR INTERVALS.
NOT AUTOMATICALLY DECLASSIFIED

MACHALT INSTRUCTION

FUEL OIL SYSTEM

MODIFICATION TO FUEL OIL CONTROL VALVE

GENERAL REGULATOR MODEL 32028

SEA 533

NOTE: BRING THIS MACHALT INSTRUCTION TO THE ATTENTION OF ALL NAVY PERSONNEL RESPONSIBLE FOR OPERATION AND MAINTENANCE OF THIS EQUIPMENT.

SEE PARAGRAPH 5 OF THIS MACHALT INSTRUCTION FOR SPECIAL APPLICABILITY REQUIREMENTS.

This document is subject to special export controls and each transmittal to foreign governments or foreign nations may only be made with prior approval of Commander, Naval Sea Systems Command, Washington, D. C. 20362.

WARNING: This material contains information affecting the national defense of the United States within the meaning of the Espionage Laws, Title 18, U.S.C., Sections 793 and 794, the transmission or revelation of which in any manner to an unauthorized person is prohibited by law.

Published by direction of Commander, Naval Sea Systems Command

CONFIDENTIAL (DATE)

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FIGURE 1. Example of classified MACHALT title page.

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MACHALT INSTRUCTION

FUEL OIL SYSTEM

MODIFICATION TO FUEL OIL CONTROL VALVE

GENERAL REGULATOR MODEL 32028

SEA 533

NOTE: BRING THIS MACHALT INSTRUCTION TO THE ATTENTION OF ALL NAVY PERSONNEL RESPONSIBLE FOR OPERATION AND MAINTENANCE OF THIS EQUIPMENT.

SEE PARAGRAPH 5 OF THIS MACHALT INSTRUCTION FOR SPECIAL APPLICABILITY REQUIREMENTS.

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FIGURE 2. Example of unclassified MACHALT title page.

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MACHALT INSTRUCTION

FUEL OIL SYSTEM

MODIFICATION TO FUEL OIL CONTROL VALVE

GENERAL REGULATOR MODEL 32028

SEA 533

NOTE: BRING THIS MACHALT INSTRUCTION TO THE ATTENTION OF ALL NAVY PERSONNEL RESPONSIBLE FOR OPERATION AND MAINTENANCE OF THIS EQUIPMENT.

SEE PARAGRAPH 5 OF THIS MACHALT INSTRUCTION FOR SPECIAL APPLICABILITY REQUIREMENTS.

NOT TO BE ACCOMPLISHED ON MATERIAL ON WHICH MACHALT _____ A HAS BEEN COMPLETED

DISTRIBUTION LIMITED TO U.S. GOVERNMENT AGENCIES ONLY. OTHER REQUESTS FOR THIS DOCUMENT MUST BE REFERRED TO COMMANDING OFFICER, NAVAL SHIP SYSTEMS ENGINEERING STATION.

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(DATE)

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FIGURE 3. Example of MACHALT revision title page.

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PREPARED BY:

(see 5.1.2.1 of standard)

IN-SERVICE ENGINEERING ACTIVITY [FOR HULL, MECHANICAL AND ELECTRICAL (HM&E) SYSTEMS.]

(see 5.1.2.2 of standard)

APPROVED FOR NAVSEA:

(see 5.1.2.3 of standard)
Signature,
Position,
Code,

Effective date

FIGURE 4. Example of MACHALT signature page.

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TITLE:

CODE IDENT 53711
NAVSEA MACHALT NO. _____ A

SUBJECT:

Date:

Approved by:

Place This Page Immediately Following the Title Page.

Revisions Record Sheet

Revision	Page

FIGURE 5. Example of MACHALT revision record sheet.

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MACHALT _____

1. DISTRIBUTION

1.1 Copies of this MACHALT instruction shall be distributed in accordance with Standard Navy Distribution List Part 1 (Edition ____ dated _____) and Part 2 (Edition ____ dated _____).

1.1.1 One copy each to the following: 24G; 24H; 26F1 (COMOPTEVFOR); 26S1; 26S2; 28K1 (COMSUBGRU 6, COMSUBDEVRON 12, COMSUBRON 2, 4, 6 and 10), 28K2 (COMSUBGRU 5, COMSUBDEVGRU 1, COMSUBRON 1, 3 and 5); 29N1 (SSN 606, 614, 615, 637, 660, 661 and 678); 29N2 (SSN 612, 647, 672 and 677); C4F53; FA13; FB13; FKA1G; (SEA 632X, SEA 634X, SEA 042, PMS 393); FKP1B (Concord, Code 3710); FKP1G (5740); FKP7 (Charleston, Mare Island, Norfolk, Pearl Harbor, Puget Sound, Portsmouth PERA (SS); FKP8 (Newport News, Pascagoula); FT38, FT46; FT54; FT64.

1.1.2 Two copies each to the following: FKA6A14 (Codes 3513, 3517, 4552, 541); FKM13 (Code 781); FKP1J.

1.1.3 Additional copies of this MACHALT can be obtained from the Naval Ship Systems Engineering Station (NAVSSES) Code 08, Philadelphia, PA 19112.

2. SUBJECT

2.1 Replacement of "O" ring seals in steering gear emergency hydraulic power unit.

3. PURPOSE

3.1 Installs "O" rings of Viton "A" throughout the steering gear emergency hydraulic power unit to prevent leakage caused by existing "O" rings.

4. AUTHORITY

4.1 Preparation of the original MACHALT was authorized by NAVSEA letter _____ dated _____ and Configuration Control Board Directive No. _____ assigned by NAVSEA to Naval Ship Systems Engineering Station (NAVSSES) Engineering Change Proposal (ECP) _____ , NSN _____ . Preparation of Revision A was authorized by NAVSEA letter _____ dated _____ and CCB Directive No. _____ Amendment _____ assigned by NAVSEA to NAVSSES ECP _____ .

FIGURE 6. Example of MACHALT page (Revision Authority).

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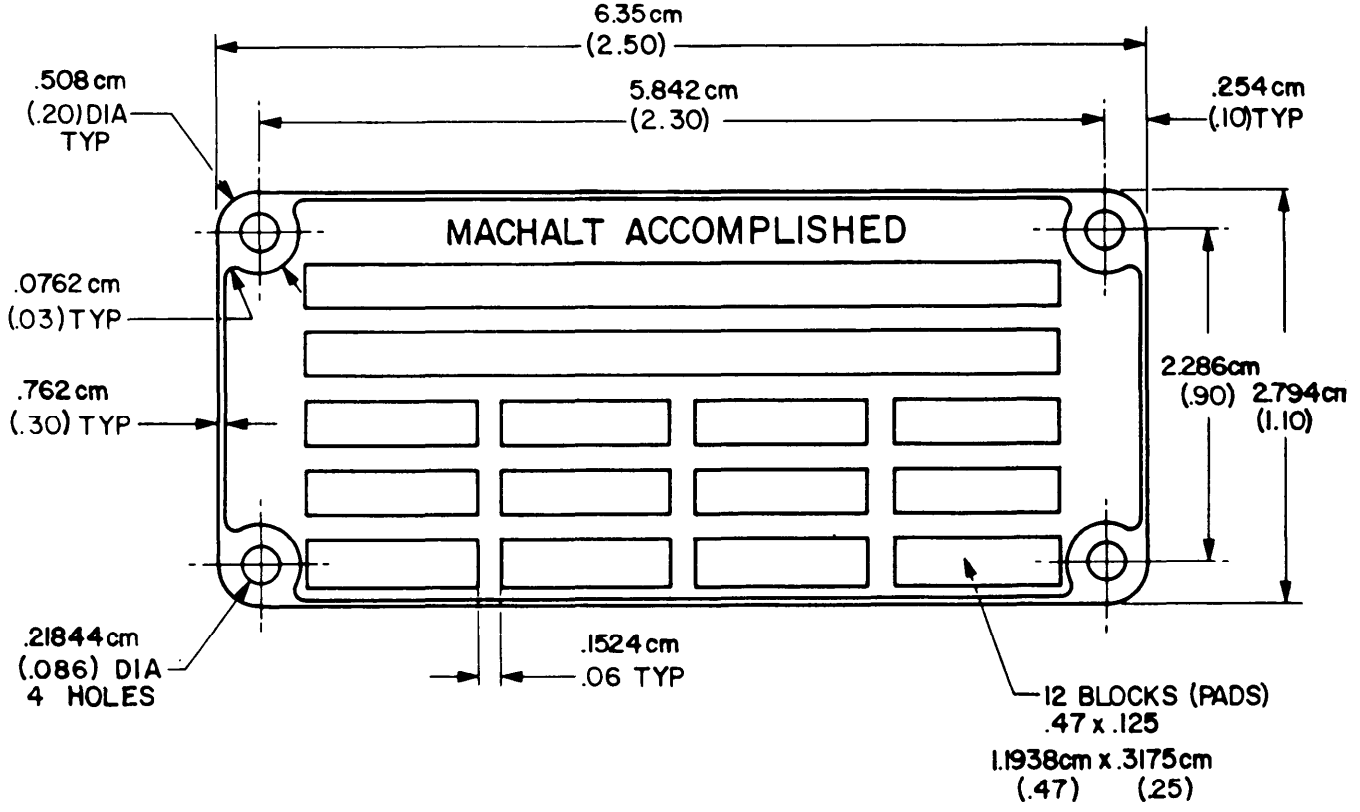


FIGURE 7. Example of MACHALT plate.

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APPENDIX A

SAMPLE MACHALT INSTRUCTION

10. SCOPE

10.1 Scope. The facsimile MACHALT instruction which follows is intended to provide personnel with a sample of a MACHALT prepared in accordance with this standard. The sample presented is for illustrative purpose only and is not an approved MACHALT.

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MACHALT _____

1. DISTRIBUTION

1.1 Copies of this MACHALT shall be distributed in accordance with the Standard Naval Distribution List [SNDLO Part 1 Edition 113 (OPNAV P09B2-107) of July 1977 and Part 2 incorporated as Section 1, Edition 4 (OPNAV P09B2-105) of January 1978].

1.1.1 One copy to each of the following ships and stations: AFS-1, AFS-2, AFS-3, AFS-7, AO-177, AO-178, AO-179, AOE-1, CG-16, CG-17, CG-18, CG-19, CG-20, CG-21, CG-22, CG-23, CG-24, CG-26, CG-27, CG-28, CG-29, CG-30, CG-31, CG-32, CG-33, CG-34, CV-59, CV-60, CV-61, CV-62, CV-63, CV-64, CV-66, CV-67, DD-931, DD-933, DD-937, DD-938, DD-940, DD-941, DD-942, DD-943, DD-944, DD-945, DD-946, DD-948, DD-950, DD-951, DDG-2, DDG-3, DDG-4, DDG-5, DDG-6, DDG-7, DDG-8, DDG-9, DDG-10, DDG-11, DDG-12, DDG-13, DDG-14, DDG-15, DDG-16, DDG-17, DDG-18, DDG-19, DDG-20, DDG-21, DDG-22, DDG-23, DDG-24, DDG-37, DDG-38, DDG-39, DDG-40, DDG-41, DDG-42, DDG-43, DDG-44, DDG-45, DDG-46, FF-1052, LHA-1, LHA-2, LHA-3, LHA-4, LHA-5, LKA-113, LKA-114, LKA-115, LKA-116, LKA-117, LPH-2, LPH-3, LPH-7, LPH-9, LPH-10, and LSD-36.

1.1.2 Two copies to the following: FKA1G, SEA 533, FKP7, (all shipyards except Mare Island) FKP1B and FKM13.

1.1.3 Six copies to the following: FKP5A

1.1.4 Fourteen copies to the following: NAVSSES [Code 023 (5), Code 041 (5), Code 08 (4)]

1.1.5 Additional copies of this instruction may be obtained from:

Commanding Officer
Naval Ship Systems Engineering Station
Philadelphia, PA 19112
Code 08

2. SUBJECT

2.1 Fuel Oil Service System; modification internal to Fuel Oil Control Valve. This MACHALT should only be accomplished on General Regulator Model 32028 valves with serial numbers preceeding 219419-79.

3. PURPOSE

3.1 This MACHALT modifies the General Regulator Model 32028 fuel oil control valve to eliminate valve internal part failure caused by fuel oil contaminants. This MACHALT also incorporates a spool material improvement to eliminate internal part failure due to corrosion.

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MACHALT _____

4. AUTHORITY

4.1 Preparation of this MACHALT was authorized by NAVSEA letter _____ dated _____ and Configuration Control Board Directive No. _____ assigned by NAVSEA to Naval Ship Systems Engineering Station (NAVSSES) Engineering Change Proposal (ECP) _____, NCN _____, UIC 65540.

5. APPLICATION

5.1 This MACHALT is applicable to General Regulator Fuel Oil Control Valves Model 32028 with serial numbers preceding 219419-79. Valves with serial number 219419-79 and higher have already been modified.

5.2 This MACHALT is applicable to Fuel Oil Control Valves located in the Fire Room in the following ships only: (Note: Quantities noted below are indicative of one valve per Main Propulsion Boiler.)

		Total Valves/Ship
USS MARS	AFS-1	3
USS SYLVANIA	AFS-2	3
USS NIAGARA FALLS	AFS-3	3
USS SAN JOSE	AFS-7	3
USS CIMARRON	AO-177	2
USS MONONGAHELA	AO-178	2
USS MERRIMAC	AO-179	2
USS SACRAMENTO	AOE-1	4
USS LEAHY	CG-16	4
USS HARRY E. YARNELL	CG-17	4
USS WORDEN	CG-18	4
USS DALE	CG-19	4
USS RICHMOND K. TURNER	CG-20	4
USS GRIDLEY	CG-21	4
USS ENGLAND	CG-22	4
USS HALSEY	CG-23	4
USS REEVES	CG-24	4
USS BELKNAP	CG-26	4
USS JOSEPHUS DANIELS	CG-27	4
USS WAINWRIGHT	CG-28	4
USS JOUETT	CG-29	4
USS HORNE	CG-30	4
USS STERETT	CG-31	4
USS WILLIAM H. STANDLEY	CG-32	4
USS FOX	CG-33	4
USS BIDDLE	CG-34	4
USS FORRESTAL	CV-59	8
USS SARATOGA	CV-60	8
USS RANGER	CV-61	8
USS INDEPENDENCE	CV-62	8
USS KITTY HAWK	CV-63	8
USS CONSTELLATION	CV-64	8
USS AMERICA	CV-66	8

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MACHALT

		Total Valves/Ship
USS JOHN F. KENNEDY	CV-67	8
USS FORREST SHERMAN	DD-931	4
USS BARRY	DD-933	4
USS DAVIS	DD-937	4
USS JONAS INGRAM	DD-938	4
USS MANLEY	DD-940	4
USS DUPONT	DD-941	4
USS BIGELOW	DD-942	4
USS BLANDY	DD-943	4
USS MULLINNIX	DD-944	4
USS HULL	DD-945	4
USS EDSON	DD-946	4
USS MORTON	DD-948	4
USS RICHARD S. EDWARDS	DD-950	4
USS TURNER JOY	DD-951	4
USS CHARLES F. ADAMS	DDG-2	4
USS JOHN KING	DDG-3	4
USS LAWRENCE	DDG-4	4
USS CLAUDE V. RICKETTS	DDG-5	4
USS BARNEY	DDG-6	4
USS HENRY B. WILSON	DDG-7	4
USS LYNDE MCCORMICK	DDG-8	4
USS TOWERS	DDG-9	4
USS SAMPSON	DDG-10	4
USS SELLERS	DDG-11	4
USS ROBISON	DDG-12	4
USS HOEL	DDG-13	4
USS BUCHANAN	DDG-14	4
USS BERKELEY	DDG-15	4
USS JOSEPH STRAUSS	DDG-16	4
USS CONYNGHAM	DDG-17	4
USS SEMMES	DDG-18	4
USS TATTNALL	DDG-19	4
USS GOLDSBOROUGH	DDG-20	4
USS COCHRANE	DDG-21	4
USS BENJAMIN STODDERT	DDG-22	4
USS RICHARD E. BYRD	DDG-23	4
USS WADDELL	DDG-24	4
USS FARRAGUT	DDG-37	4
USS LUCE	DDG-38	4
USS MACDONOUGH	DDG-39	4
USS COONTZ	DDG-40	4
USS KING	DDG-41	4
USS MAHAN	DDG-42	4
USS DAHLGREN	DDG-43	4
USS WILLIAM V. PRATT	DDG-44	4
USS DEWEY	DDG-45	4
USS PREBLE	DDG-46	4
USS KNOX	FF-1052	2
USS TARAWA	LHA-1	2
USS SIAPAN	LHA-2	2

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MACHALT

		Total Valves/Ship
USS BELLEAU WOOD	LHA-3	2
USS NASSAU	LHA-4	2
USS PELELIU	LHA-5	2
USS CHARLESTON	LKA-113	2
USS DURHAM	LKA-114	2
USS MOBILE	LKA-115	2
USS ST. LOUIS	LKA-116	2
USS EL PASO	LKA-117	2
USS IWO JIMA	LPH-2	2
USS OKINAWA	LPH-3	2
USS GUADALCANAL	LPH-7	2
USS GUAM	LPH-9	2
USS TRIPOLI	LPH-10	2
USS ANCHORAGE	LSD-36	2

6. MACHALT ACCOMPLISHMENT KEYPOINT CHECK

6.1 Verification of MACHALT accomplishment is required. The following keypoint check shall be made:

6.1.1 Ensure that MACHALT Plate is secured to the valve body of the Fuel Oil Control Valve.

7. COMPLIANCE

7.1 This MACHALT shall be accomplished on those Fuel Oil Control Valves located in ships identified in paragraph 5 in accordance with installation priorities established by the CNO Amalgamated Military/ Technical Improvement Plan (AMT) based on an assigned OPNAV MACHALT priority Level #2, Reliability and Maintainability Primary System Modernization.

8. ACCOMPLISHMENT CATEGORY

8.1 This is a category 2 (intermediate) MACHALT and shall be accomplished during tender/shore base availability by Naval Sea Support Center personnel or by tender/shore base personnel under direction or supervision of Naval Sea Support Center personnel.

9. MAN-HOURS REQUIRED

9.1 It is estimated that a total of 15 man-hours will be required to accomplish the work required by this MACHALT.

10. DRAWING REFERENCES

NOTE: Shipboard microfilm aperature card files shall be maintained by inserting cards for new drawing supplied with the MACHALT kit. Revised drawings will be forwarded separately by delegated agents.

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MACHALT _____

10.1 New Drawings

10.1.1 (53711) 5783211

10.2 Drawings Requiring Revision

10.2.1

(53711) 5761600	Sheets 1, 2	Obtain from Norfolk Naval Shipyard

10.2.2

(10001) 2875618-S	Sheets 1, 3	Obtain from Long Beach Naval Shipyard

10.2.3

(10001) 2876208-T	All Sheets	

10.3 Obsolete Drawings

10.3.1

(10001) 2751321	All Sheets	Destroy

10.4 For Information Only.

10.4.1 (10001) 2662548-A

11. ENCLOSURE

11.1 Supply/Material Support Data

12. SUPPLY DATA

12.1 MACHALT kit should be ordered from Commanding Officer, Naval Ship Systems Engineering Station (NAVSSES), Code 002, Philadelphia, PA 19112. Order MACHALT kit number _____. Each MACHALT kit includes installation and checkout material, logistic support products and documentation required to alter one Fuel Oil Control Valve.

12.2 Kit Contents

12.2.1 Each MACHALT kit includes two copies of this MACHALT instruction and the following:

12.2.1.1 Drawings listed in paragraph 10.1.

12.2.1.2 Updated support documentation listed paragraph 16.1.

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MACHALT _____

12.2.1.3 Materials

ITEM NO.	NOMENCLATURE/ DESCRIPTION	NATIONAL STOCK NO.	PART NO.	MEC CODE	SMR CODE	MACHALT KIT QUANTITY
1	"O" Ring, General Regulator	5407-00-521-7711	X-73119-00	1	PA6XX	8
2	Bellows Assembly, General Regulator	5310-00-516-7821	50508-01	1	PB7ZZ	4
3	"O" Ring, General Regulator	5407-00-521-7710	X-72375-01	1	PB3BB	4
4	Stop Collar, General Regulator	5306-00-616-5121	32609-01	1	PA5XX	4
5	Gasket, General Regulator	5100-00-360-2500	B-32608-01	1	PC3AB	4
6	Sleeve & Stem Assembly	5210-00-521-6000	D-32612-02	1	PA8UR	1

12.2.1.4 Identification Plate

12.3 Parts/Materials/Special Tools or Test Equipment required for accomplishment but not supplied in Kit.

12.3.1 None

12.4 Special Tools/Support/Test Equipment required after MACHALT accomplishment:

12.4.1 None

12.5 Disposition of Removed Material/Parts and/or Units

Quantity	Part No.	Nomenclature	National Stock No.	Disposition Code
4	50508-01	Bellows Assembly, General Regulator	5310-00-516-7821	2
4	32028-00	Stop Collar, General Regulator	5306-00-616-5121	2
1	32612-01	Sleeve and Stem Assembly	5210-00-521-6000	2

12.6 Alteration of Spare Assemblies/Equipment.

12.6.1 None

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MACHALT _____

12.7 Parts/Materials Required to Modify Spares.

12.7.1 None

12.8 New/Modified Additional Spare Parts

12.8.1

Item No.	Nomenclature/ Description	National Stock No.	Part No.	Quantity
1	"O" Ring, General Regulator	5407-00-521-7711	X-73119-00	4
2	"O" Ring, General Regulator	5407-00-521-7710	X-72375-01	2
3	"O" Ring, General Regulator	5407-00-521-7709	X-75982-01	2

12.9 Allowance Changes

12.9.1 The following items/quantities are to be added to the present Allowance Parts List (APL) after equipment is altered:

Part No.	Nomenclature	National. Stock No.	SM&R Code	Parts Population Added	Allowance Quantity Increase
X-73119-00	"O" Ring, General Regulator	5407-00-521-7711	1	6	6
X-72375-01	"O" Ring, General Regulator	5407-00-521-7710	1	6	6
X-75982-01	"O" Ring, General Regulator	5407-00-521-7709	1	6	6

12.9.2 The following items/quantities are deleted from the present repair parts allowance (COSAL) after equipment is altered:

12.9.2.1 None

13. DETAILED INSTRUCTIONS

13.1 Disassembly

The following is a complete disassembly procedure for the control valve. If a specific component requires removal, it is only necessary to proceed through those steps required to remove that part. Disassemble the control valve in the following manner. (Refer to Figure 8).

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MACHALT

13.1.1 Close manual isolation valves on either side of control valve and shut off signal air. Disconnect input signal airline from valve.

13.1.2 Loosen bellows chamber bleed valve and bleed off trapped fuel pressure; then disconnect connector (21) and remove tubing (19).

13.1.3 Turn handwheel completely counterclockwise.

13.1.4 Loosen jam nut (1) and loosen spring adjusting screw (30).

13.1.5 Remove nuts (6) and capscrews (3).

13.1.6 Remove spring chamber (2) and loading spring (29).

13.1.7 Remove piston (28), diaphragm (27), spacer plate (5), and screws (4) as an assembly.

13.1.8 Remove seal ring (26).

13.1.9 Remove piston (25), diaphragm (27), spacer plate (5), screws (4), nut 24 and pushrod (20) as an assembly.

13.1.10 Remove nuts (12).

13.1.11 Pull bellows chamber (10), with seat, straight up and away from valve body. Remove and discard gasket (14).

13.1.12 Open bleed valve (8) and turn assembly sideways to drain fuel from bellows chamber.

13.1.13 Insert screwdriver in lower slotted projection of bellows assembly (9) and, while holding bellows assembly tight to prevent twisting, loosen and remove stem (16) using a 5/32 inch diameter rod inserted in lower stem hole.

13.1.14 Loosen and remove screws (23) and lockwashers (22); then remove clamp plates (7).

CAUTION: When removing bellows, never push it from below as this causes distortion of the bellows.

13.1.15 Using puller and long screws inserted in threaded holes of upper bellows end, remove bellows (9) and "O" ring (13). Discard "O" ring.

13.1.16 Remove lockwasher (11) and stop collar (18).

13.1.17 Using appropriate size wrench, remove sleeve (17). Remove and discard "O" rings (13 and 15).

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13.2 Assembly

The following is a complete assembly procedure for the control valve. If only a specific component was removed, proceed to the step that describes installation of that part and continue with the procedure. Assemble the components of the valve in the following manner. Refer to Figure 8.

13.2.1 Install sleeve (17), using appropriate size wrench, with new "O" rings (13 and 15).

13.2.2 Insert stem (16) in sleeve (17).

13.2.3 Install stop collar (18) and lockwasher (11).

13.2.4 Install bellows (9) and new "O" ring (13).

13.2.5 Install clamp plates (7) and lockwashers (22). Tighten screws (23).

13.2.6 Insert screwdriver in lower slotted projection of bellows assembly (9) and, while holding bellows assembly tight to prevent twisting, install and tighten stem (16) using a 5/32 inch diameter rod inserted in lower stem hole.

13.2.7 Close bleed valve.

13.2.8 Using a new gasket (14), install bellows chamber (10), with seat, in valve body.

13.2.9 Install and tighten nuts (12).

13.2.10 Install piston (25), diaphragm (27), spacer plate (5), screws (4), nut (24), and pushrod (20) as an assembly.

13.2.11 With valve completely open and stop collar (18) seated against sleeve (17), adjust pushrod (20) in piston (25) until distance between point A and point B is 0.375 inch. Secure pushrod by tightening nut (24).

13.2.12 Install seal ring (26).

13.2.13 Install piston (28), diaphragm (27), spacer plate (5), and screws (4) as an assembly.

13.2.14 Install loading spring (29) and spring chamber (2).

13.2.15 Install capscrews (3) and nuts (6).

13.2.16 Install tubing (19) and fitting (21).

13.2.17 Connect signal airline to valve, apply signal air, and open isolation valves on either side of control valve.

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MACHALT _____

13.3 Adjustment

13.3.1 Apply fuel to valve and purge bellows chamber of air and residual water by slowly opening upstream manual shutoff valve while bleed screw in vent valve in side of bellows chamber is open until fuel trickles from vent. When fuel begins to flow through vent, tighten bleed screw in place in vent.

13.3.2 Minimum fuel pressure adjustment should be accomplished as follows:

CAUTION: Do not allow steam pressure to decrease more than 50 PSI below setpoint (1200 PSI ships) or more than 25 PSI below setpoint (600 PSI ships). Place forced draft blower controls in remote or local manual. Observe stack condition at all times and adjust combustion air flow to prevent smoking.

13.3.2.1 Transfer control of fuel pressure to local manual and vent control air signal from the valve.

13.3.2.2 Place as many burners in service as the boiler load can accommodate.

13.3.2.3 Manually decrease fuel pressure by turning the handwheel of the valve. Do not allow fuel pressure to drop below the specified minimum. Rotate the handwheel to the end of its travel or until minimum fuel pressure is established.

13.3.2.4 Loosen the jam nut at the base of the adjusting screw and turn the adjusting screw until the desired fuel pressure is established. If the handwheel was not at the end of its travel, turn the adjusting screw clockwise until fuel pressure increases slightly above minimum, then rotate the handwheel to the end of its travel and set the fuel pressure at the desired minimum by turning the adjusting screw counterclockwise. Lock the final setting by tightening the jam nut while holding the adjusting screw stationary.

13.4 Key to Figure 8

1 Jam nut
2 Spring Chamber
3 Capscrew
4 Screw
5 Spacer Plate
6 Nut
7 Clamp Plate
8 Bleed Valve
9 Bellows Assembly
10 Bellows Chamber

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MACHALT _____

11 Lockwasher
12 Nut
13 "O" Ring
14 Gasket
15 "O" Ring
16 Stem
17 Sleeve
18 Stop Collar
19 Tubing
20 Pushrod
21 Connector
22 Lockwasher
23 Screw
24 Nut
25 Piston
26 Seal Ring
27 Diaphragm
28 Piston
29 Loading Spring
30 Spring Adjusting Screw

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MACHALT

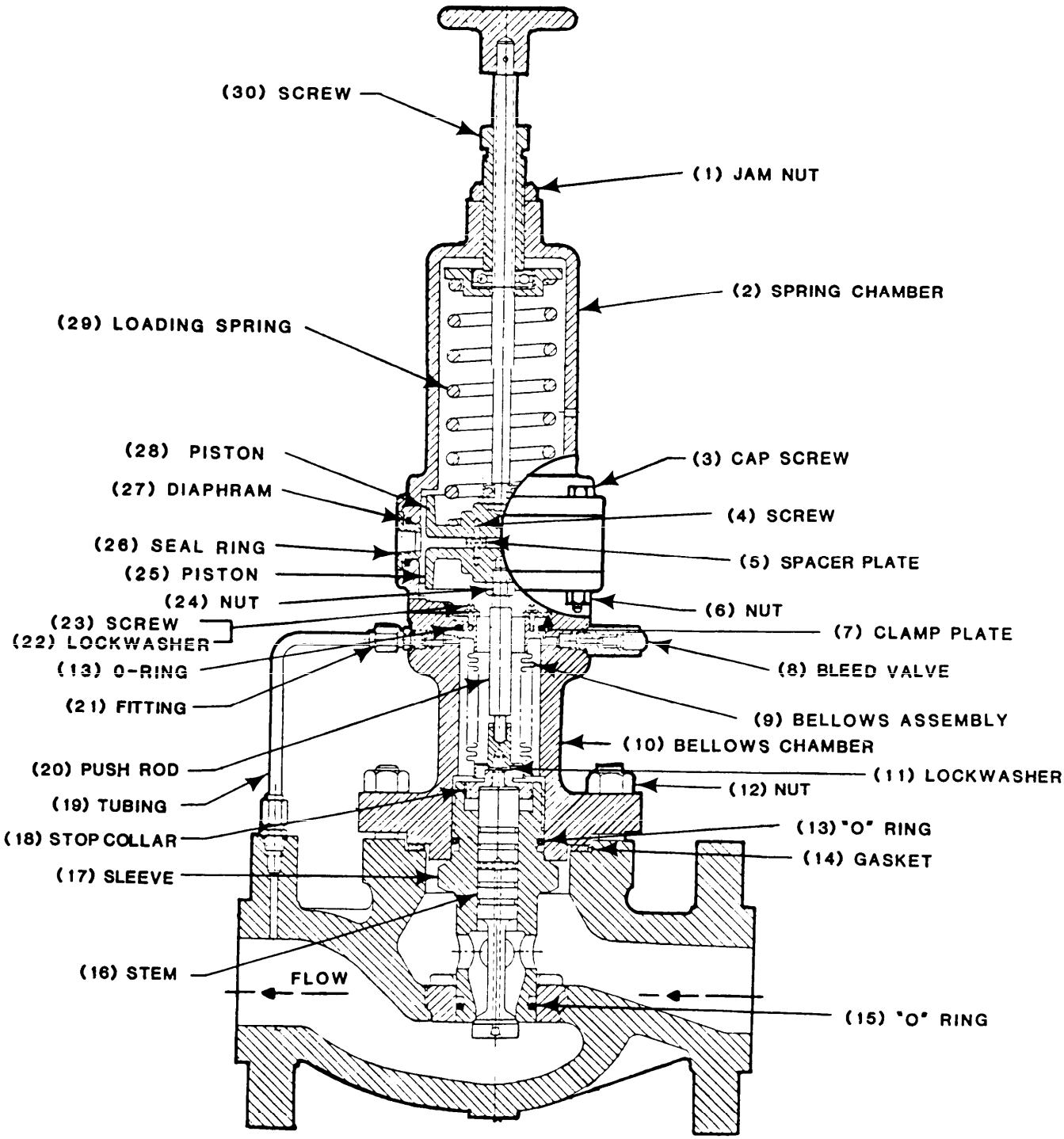


FIGURE 8. Cutaway view of valve assembly.

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MACHALT _____

14. IDENTIFICATION

14.1 Install Identification Plate by safety wiring to the valve body after accomplishment of the MACHALT.

15. SHIPPING WEIGHT

15.1 MACHALT kit consists of one package with a volume of .42 cu. ft., size 12" by 10" by 6", and weight of approximately 19 pounds.

16. UPDATING OPERATIONAL SUPPORT DOCUMENTATION

16.1 The following publication changes are required for system/ equipment support following MACHALT accomplishment.

16.1.1 Technical Manual

16.1.1.1 NAVSEA 9000-LP-000-0000

16.1.2 Maintenance Index Pages/Requirements Cards (MIPs/MRCs).

16.1.2.1 MIP _____ , MRC _____

16.1.3 Allowance Parts List (APL).

16.1.3.1 _____

16.1.3.2 Interim support of this MACHALT is provided by MACHALT APL _____. When all installations are complete, the parent APL listed in 16.1.3.1 will be updated in accordance with paragraph 12.9.1.

16.1.4 Equipment Identification Codes (EICs).

16.1.4.1 The following EIC shall be used when reporting MACHALT accomplishment on OPNAV Form 4790/2K, as required by paragraph 17.2

16.1.4.2 EIC _____

16.1.5 Technical Repair Standards (TRSs).

16.1.5.1 None

16.1.6 Total Ship Test Procedures (TSTPs).

16.1.6.1 None

16.1.7 Engineering Operational Sequencing System (EOSS)

16.1.7.1 None

16.1.8 Other Data

16.1.8.1 None

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16.2 Documentation Availability

16.2.1 The updated support documentation specified in paragraph 16.1 is made available as part of the MACHALT kit. Automatic distribution of subsequent maintenance documentation to the Command will be accomplished by submission of OPNAV Form 4790/CK. This material shall not be used to support the Fuel Oil Service System or Fuel Oil Control Valve until the MACHALT has been accomplished.

16.3 Drawing Availability

16.3.1 Drawings required to support this MACHALT are specified in paragraphs 10.1 and 10.2.

17. REPORT OF COMPLETION/LOG ENTRY

17.1 The MACHALT Record Plate shall be installed in accordance with paragraph 14.1 upon completion of this MACHALT.

17.2 MACHALT accomplishment shall be entered on the Maintenance Data Form, OPNAV Form 4790/CK in accordance with OPNAVINST 4790.4.

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MACHALT _____

ENCLOSURE 11.1

SUPPLY/MATERIAL SUPPORT DATA

I. PURPOSE

1. This list is for use by Ship's Supply Officer for correcting and updated onboard APL/COSAL records.

II. SUPPLY DATA

1. The format and content of the following is identical to paragraph 12.9 of the basic MACHALT instruction:

12.9 Allowance Changes.

12.9.1 The following items/quantities are to be added to the present Allowance Parts List (APL) after equipment is altered:

Part No.	Nomenclature	National Stock No.	SM&R Code	Parts Population Added	Allowance Quantity Increase
X-73119-00	"O" Ring, General Regulator	5407-00-521-7711	1	6	6
X-72375-01	"O" Ring, General Regulator	5407-00-521-7710	1	6	6
X-75982-01	"O" Ring, General Regulator	5407-00-521-7709	1	6	6

12.9.2 The following items/quantities are deleted from the present COSAL after equipment is altered:

12.9.2.1 None

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APPENDIX B

DATA REQUIREMENTS

10. DATA

10.1 Data requirements. When this standard is used in a contract which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this standard is cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1	Machinery alteration instruction	DI-E-2188	------

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

10.1.1 The data requirements of 10.1 and any task in the standard required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this standard. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

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Characteristics identified in 4.5.4 shall have an AQQL of 2.0%. All other

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
DOD-STD-2140(SH) Metric

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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